

Transmission Network Modelling & Flow-Based Market Coupling in SDDP

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Our team here today



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Tools

Transmission Network Modelling in SDDP

Overview

Transmission Modelling in SDDP

Alternatives for representing transmission aspects:



Zonal

Linearized Power Flow

- ✓ Represents energy transfer limits between neighboring systems
- ✓ Considers explicit linear losses through loss factors



Full Nodal Representation

Linearized Power Flow

- ✓ Represents **first** and **second Kirchhoff laws**, as well as **power flow limits** on individual circuits
- ✓ Represents quadratic losses
- ✓ Represents DC links

Flow-Based Market Coupling

DC Power Flow

Nodal Representation

First Kirchhoff Law

$$\sum_{j \in \Omega_i} (f_{ij} - f_{ji}) + g_i = d_i \quad \text{for } i = 1, \dots, N$$

$$\underbrace{\begin{pmatrix} \vdots & \vdots \\ S_{ij} & \vdots \\ \vdots & \vdots \end{pmatrix}}_{N \times M} \underbrace{\begin{pmatrix} \vdots \\ f_{ij} \\ \vdots \end{pmatrix}}_{M \times 1} + \underbrace{\begin{pmatrix} \vdots \\ g_i \\ \vdots \end{pmatrix}}_{N \times 1} = \underbrace{\begin{pmatrix} \vdots \\ d_i \\ \vdots \end{pmatrix}}_{N \times 1}$$

$$\gamma_m = \frac{1}{x_m}$$

Second Kirchhoff Law

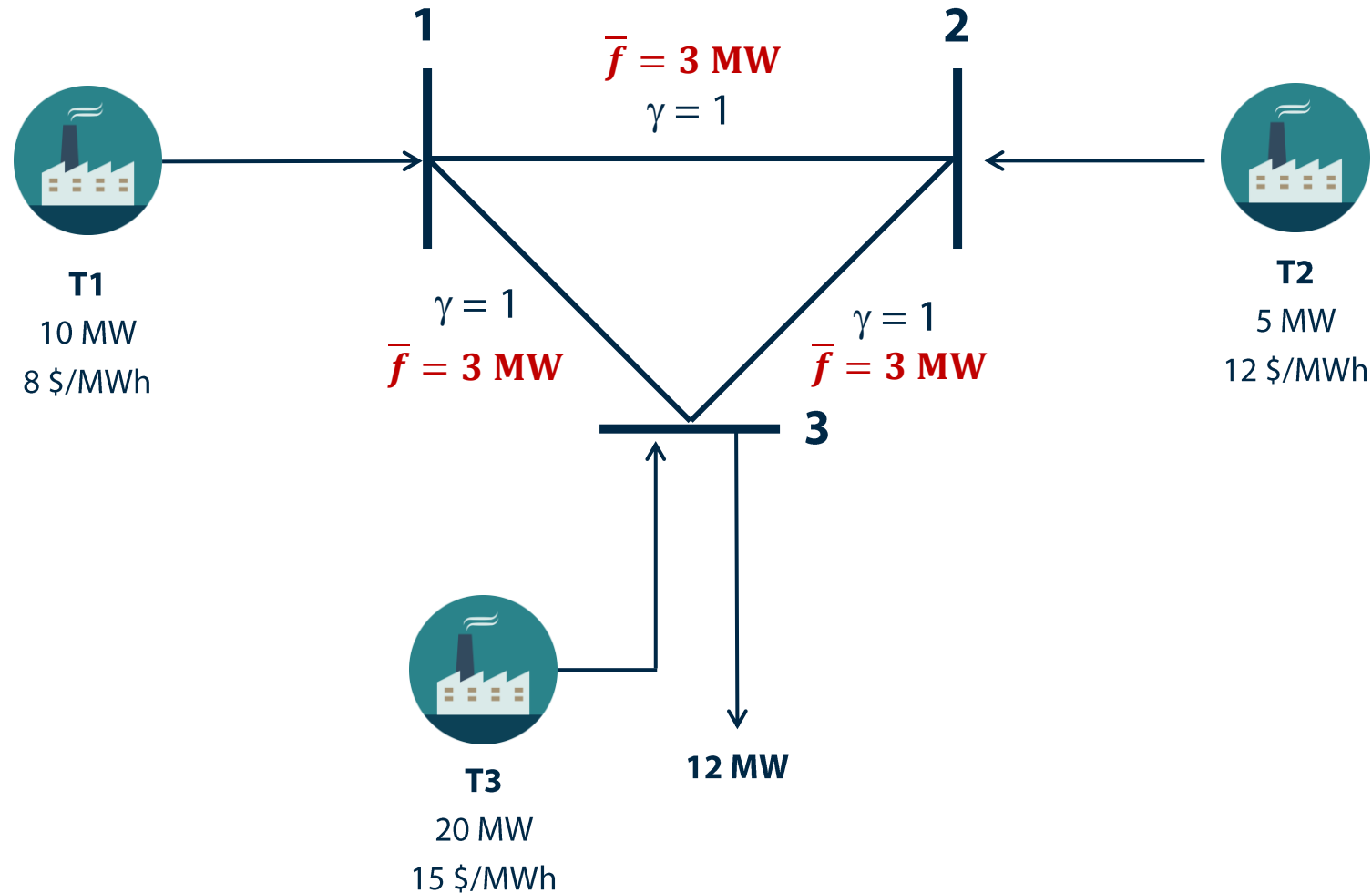
$$f_m = \gamma_m \cdot (\theta_i - \theta_j) \quad \text{for } m = 1, \dots, M$$

$$\underbrace{\begin{pmatrix} \vdots \\ f_{ij} \\ \vdots \end{pmatrix}}_{M \times 1} = - \underbrace{\begin{pmatrix} \triangle & 0 \\ \gamma_m & \triangle \\ 0 & \triangle \end{pmatrix}}_{M \times M} \underbrace{\begin{pmatrix} \vdots \\ S_{ji} \\ \vdots \end{pmatrix}}_{M \times N} \underbrace{\begin{pmatrix} \vdots \\ \theta_i \\ \vdots \end{pmatrix}}_{N \times 1}$$

Circuit Flow Limits

$$-\bar{f}_m \leq f_m \leq \bar{f}_m \quad \text{for } m = 1, \dots, M$$

Example 1 – Base case



Example 1 – Formulation with DC OPF Equations

$$\text{Min } 8g_1 + 12g_2 + 15g_3$$

Subject to:

$$g_1 - f_{12} - f_{13} = 0 \text{ MW}$$

$$g_2 + f_{12} - f_{23} = 0 \text{ MW}$$

$$g_3 + f_{13} + f_{23} = 12 \text{ MW}$$

$$f_{12} = 1 * \Delta\theta_{12}$$

$$f_{13} = 1 * \Delta\theta_{13}$$

$$f_{23} = 1 * \Delta\theta_{23}$$

$$g_1 \leq 10 \text{ MW}$$

$$g_2 \leq 05 \text{ MW}$$

$$g_3 \leq 20 \text{ MW}$$

$$\theta_1 = \text{free}$$

$$\theta_2 = \text{free}$$

$$\theta_3 = 0$$

$$-3 \leq f_{12} \leq +3$$

$$-3 \leq f_{13} \leq +3$$

$$-3 \leq f_{23} \leq +3$$

PTDF Formulation – Removing f and θ variables

DC Power Flow

Replacing $f = -[\gamma] S^t \theta$ in the Kirchhoff's 1st law

$$S f + g = d \quad \Rightarrow \quad -S [\gamma] S^t \theta + g = d$$

Let $B = -S[\gamma]S^t$, be the susceptance matrix:

$$B \theta + g = d \quad \Rightarrow \quad \theta = B^{-1} (d - g)$$

Replacing $\theta = B^{-1} (d - g)$ in the Kirchhoff's 2nd law

$$f = -[\gamma] S^t \theta \quad \Rightarrow \quad f = -[\gamma] S^t B^{-1} (d - g)$$

Let $PTDF = [\gamma] S^t B^{-1}$ be the sensitivity matrix:

$$f = PTDF \times \underbrace{(g - d)}_{\text{net position}}$$

$$\begin{array}{ll} \text{Min} & c^t g \\ \text{S.t.} & e^t g = e^t d \\ & g \leq g_{\max} \\ & -f_{\max} \leq PTDF \times (g - d) \leq f_{\max} \end{array}$$

Example 1 – PTDF Formulation

Formulation

- Incidence matrix bus-circuit:

$$S = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{transposed} \quad S^t = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

- Susceptance matrix:

$$B = -S\gamma S^t = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \quad \gamma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Modified susceptance matrix (B3 as reference bus)

$$B^t = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & 10^{30} \end{bmatrix} \quad \text{inverse} \quad (B^t)^{-1} = \begin{bmatrix} -0.66 & -0.33 & 10^{-30} \\ -0.33 & -0.66 & 10^{-30} \\ 10^{-30} & 10^{-30} & 10^{-30} \end{bmatrix}$$

- Sensitivity matrix:

$$\beta = -\gamma S^t (B^t)^{-1} = \begin{bmatrix} -0.33 & 0.33 & 0 \\ -0.66 & -0.33 & 0 \\ -0.33 & -0.66 & 0 \end{bmatrix}$$

Example 1 – Explicit formulation (DC Power Flow)

$$\text{Min } 8g_1 + 12g_2 + 15g_3$$

Subject to:

$$g_1 - f_{12} - f_{13} = 0 \text{ MW}$$

$$g_2 + f_{12} - f_{23} = 0 \text{ MW}$$

$$g_3 + f_{13} + f_{23} = 12 \text{ MW}$$

$$f_{12} = 1 * \Delta\theta_{12}$$

$$f_{13} = 1 * \Delta\theta_{13}$$

$$f_{23} = 1 * \Delta\theta_{23}$$

$$g_1 \leq 10 \text{ MW}$$

$$g_2 \leq 05 \text{ MW}$$

$$g_3 \leq 20 \text{ MW}$$

$$\theta_{12} = 0$$

$$\theta_{13} = \text{free}$$

$$\theta_{12} = \text{free}$$

$$-10 \leq f_{12} \leq +10$$

$$-3 \leq f_{13} \leq +3$$

$$-10 \leq f_{23} \leq +10$$

Example 1 – Formulation with PTDFs

$$\text{Min } 8g_1 + 12g_2 + 15g_3$$

Subject to:

$$g_1 + g_2 + g_3 = 12 \text{ MW}$$

$$-10 \leq +0.333 g_1 - 0.333 g_2 - 0 * (12 - g_3) \leq +10$$

$$-3 \leq +0.667 g_1 + 0.333 g_2 - 0 * (12 - g_3) \leq +3$$

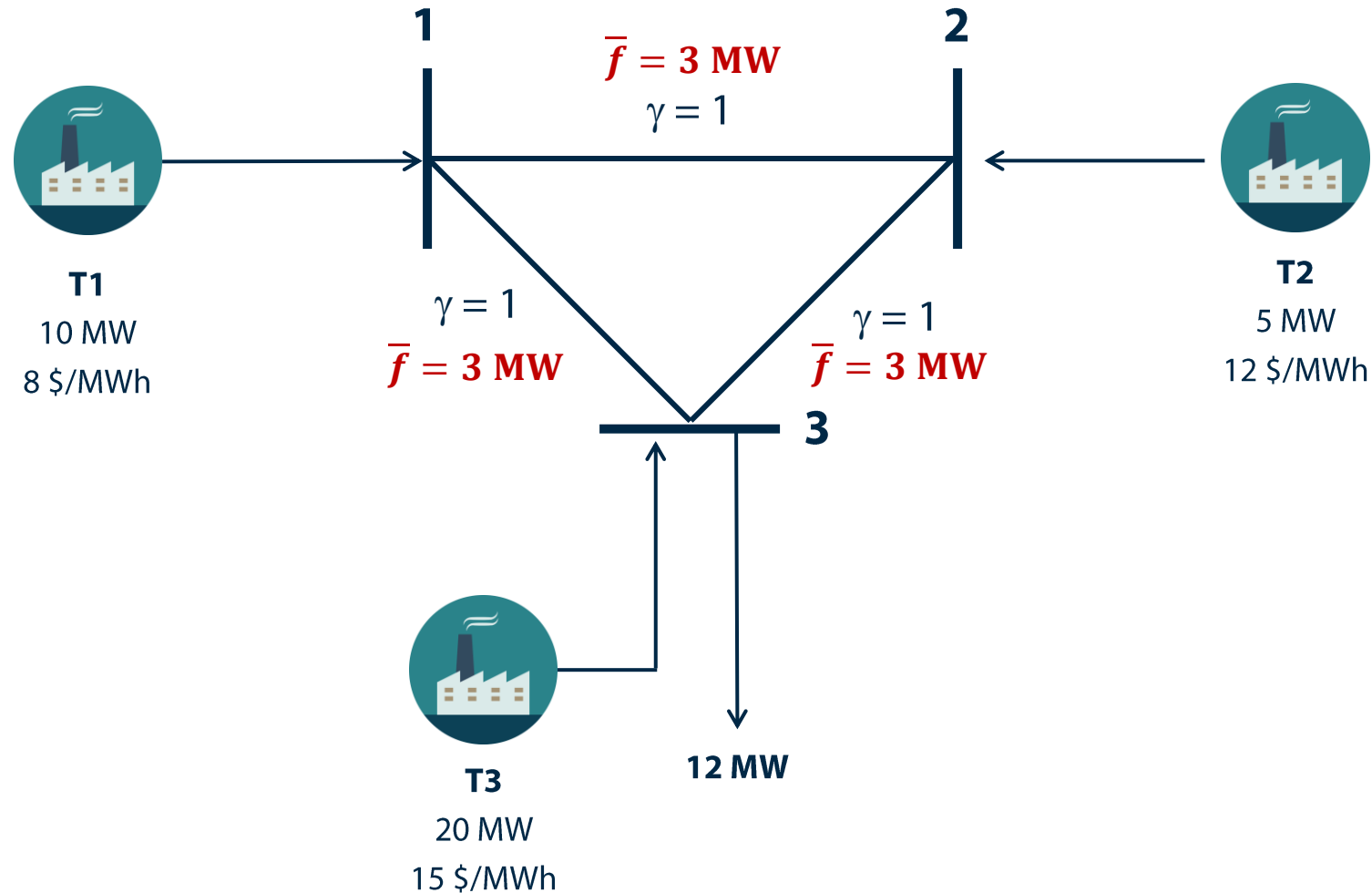
$$-10 \leq +0.333 g_1 + 0.667 g_2 - 0 * (12 - g_3) \leq +10$$

$$g_1 \leq 10 \text{ MW}$$

$$g_2 \leq 05 \text{ MW}$$

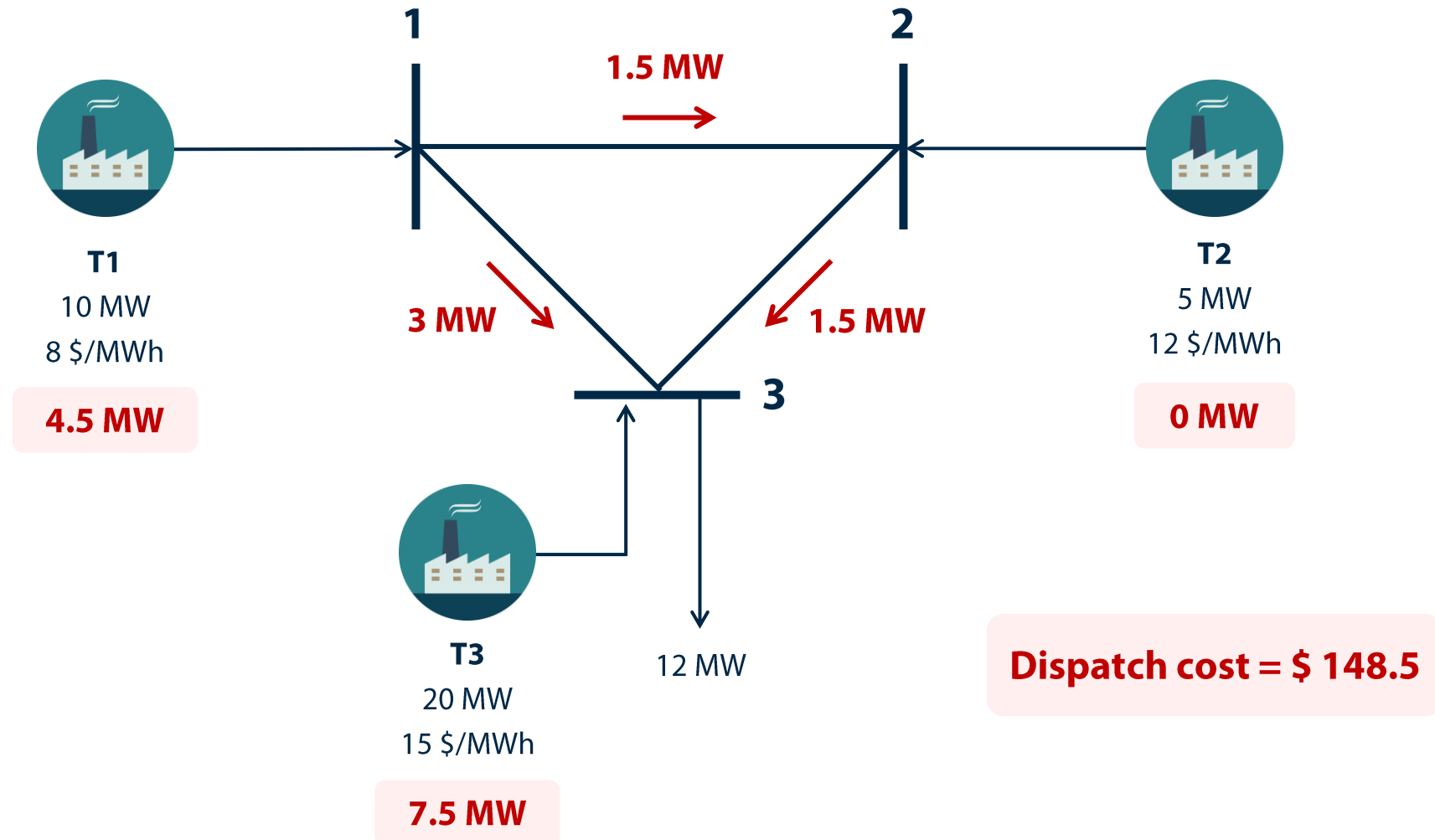
$$g_3 \leq 20 \text{ MW}$$

Example 1 – Base case (recalling...)



Example 1 – Optimal Solution

Capacity limit of 3 MW on circuit 1-3





Security-Constrained Optimal Power Flow (SCOPF)

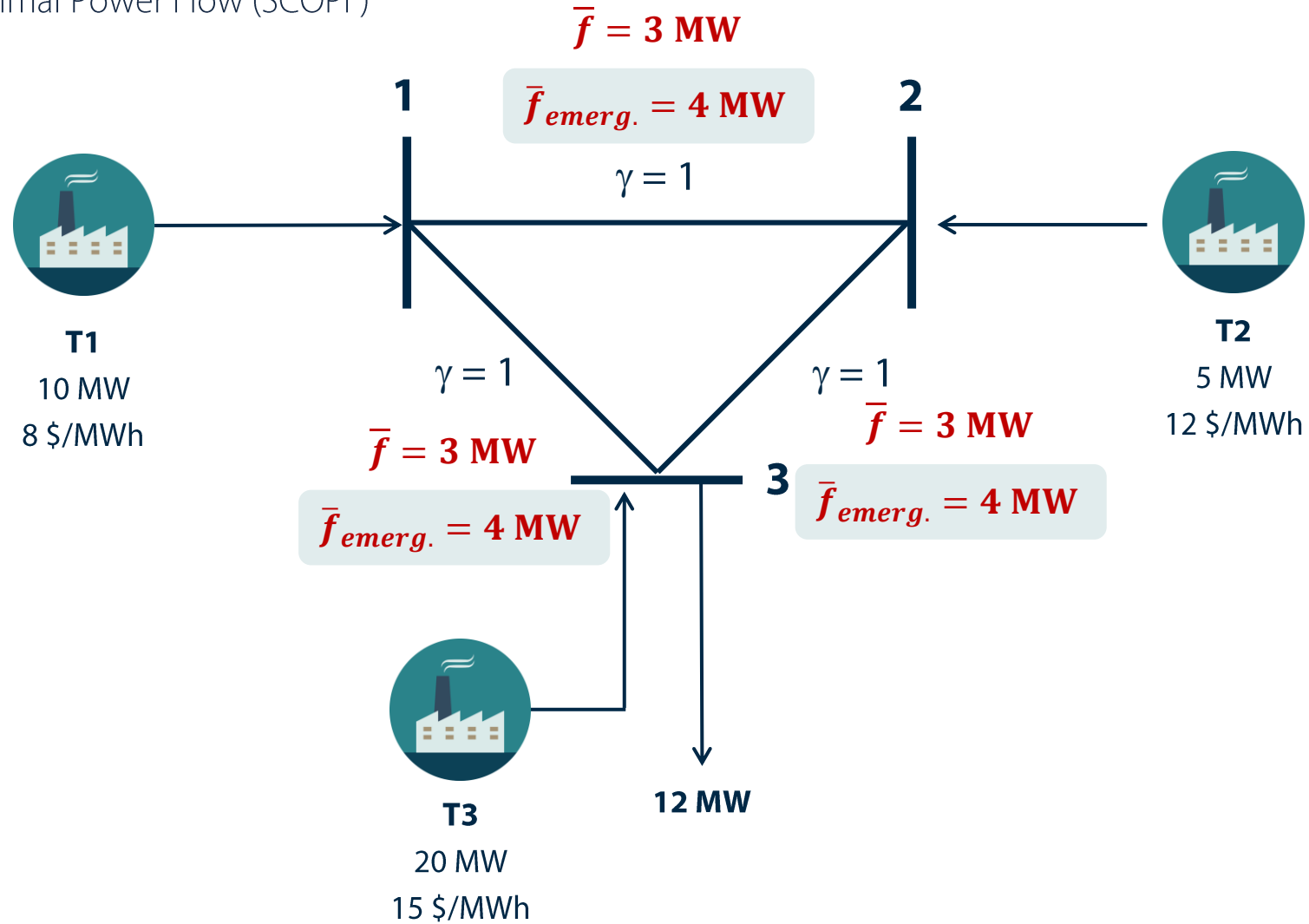
Security Constraints (N-1)

Security-Constrained Optimal Power Flow (SCOPF)

- **Objective:** Ensure the dispatch is secure under single contingencies.
- **How:** Flows on monitored circuits must remain below their emergency limits even after a component outage (CNECO).
- **Scope:** Typically applied to single-circuit failures (N-1).

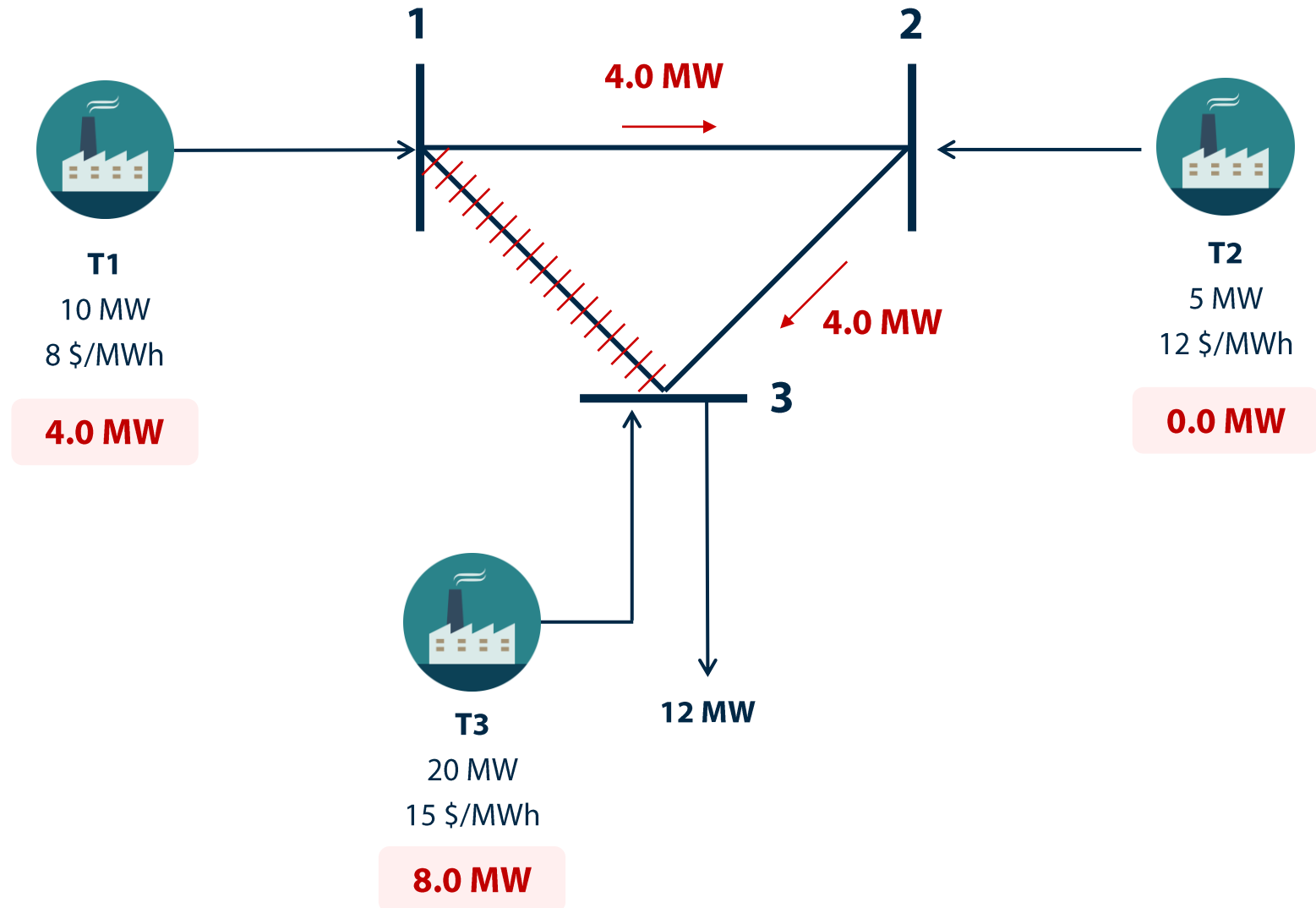
Example: Base Case

Security-Constrained Optimal Power Flow (SCOPF)



Example: Contingency on the Circuit 1-3

Security-Constrained Optimal Power Flow (SCOPF)



Example 1 – Formulation with PTDFs

$$\text{Min } 8g_1 + 12g_2 + 15g_3$$

Subject to:

$$g_1 + g_2 + g_3 = 12 \text{ MW}$$

$$-3 \leq +0.333 g_1 - 0.333 g_2 - 0 * (12 - g_3) \leq +3$$

$$-3 \leq +0.667 g_1 + 0.333 g_2 - 0 * (12 - g_3) \leq +3$$

$$-3 \leq +0.333 g_1 + 0.667 g_2 - 0 * (12 - g_3) \leq +3$$

$$+ 1.000 g_1 + 0 g_2 - 0 * (12 - g_3) \leq +4$$

$$+ 1.000 g_1 + 1.000 g_2 - 0 * (12 - g_3) \leq +4$$

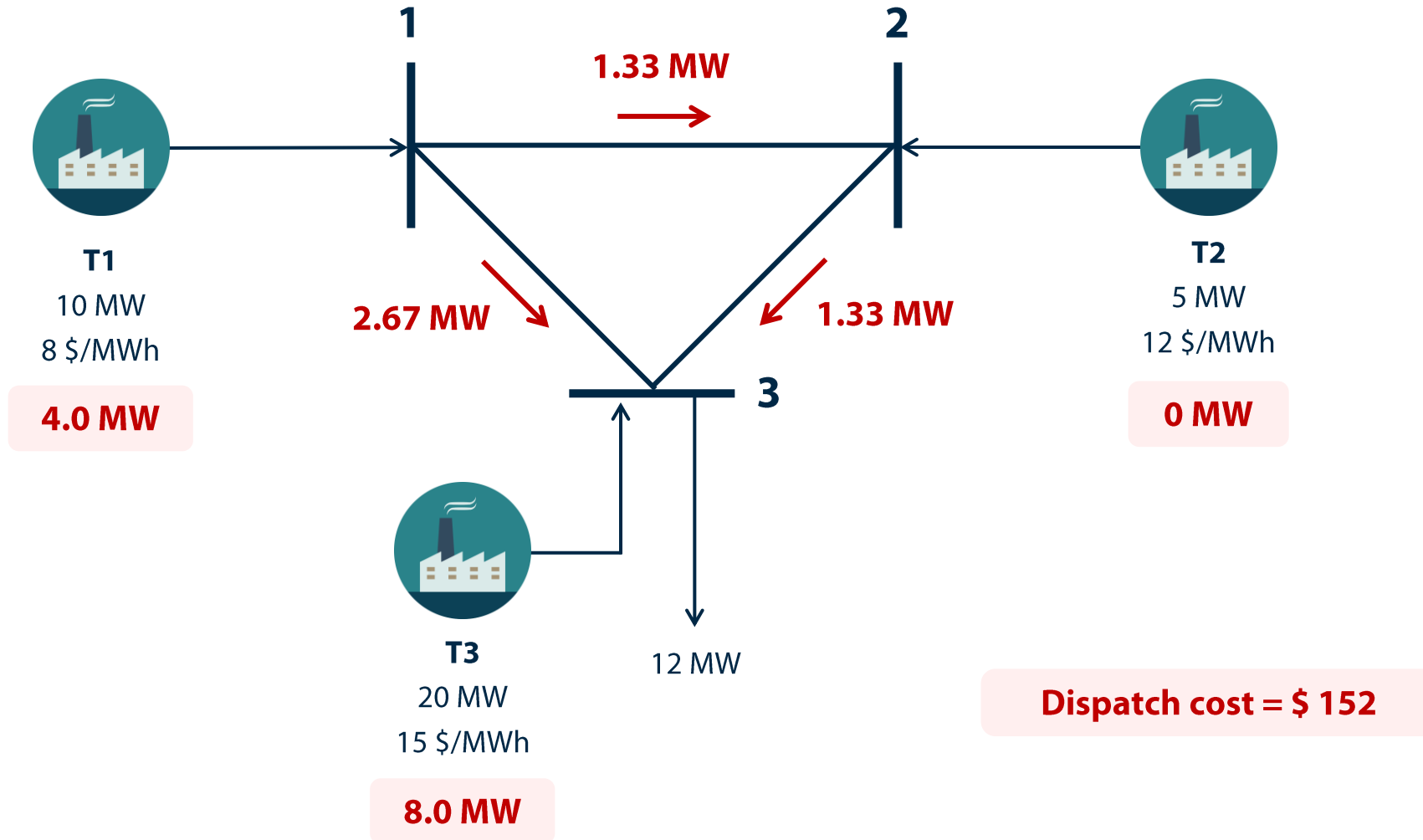
$$g_1 \leq 10 \text{ MW}$$

$$g_2 \leq 05 \text{ MW}$$

$$g_3 \leq 20 \text{ MW}$$

Example: Optimal Solution

Security-Constrained Optimal Power Flow (SCOPF)

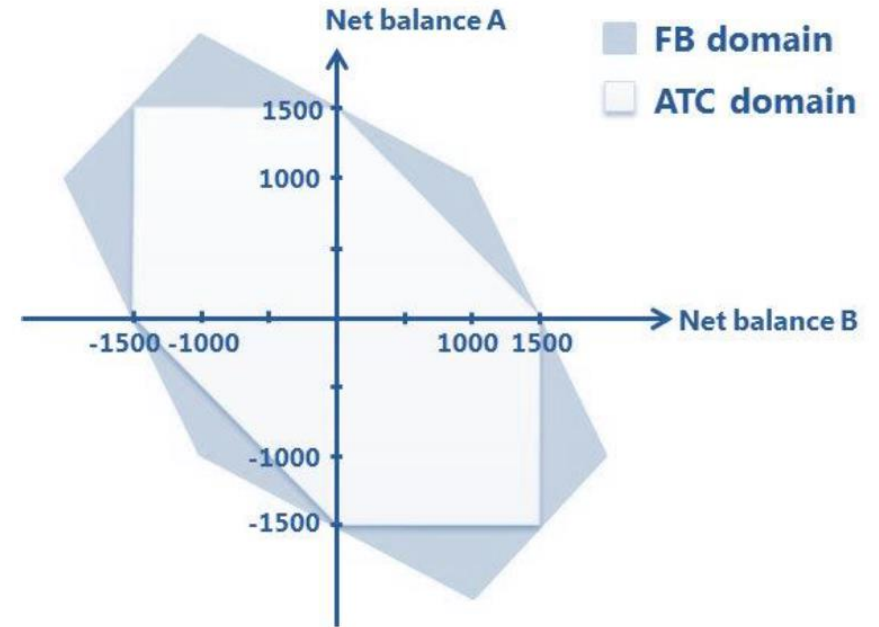




Flow-Based Market Coupling (FBMC)

Flow-Based Market Coupling (FBMC) Overview

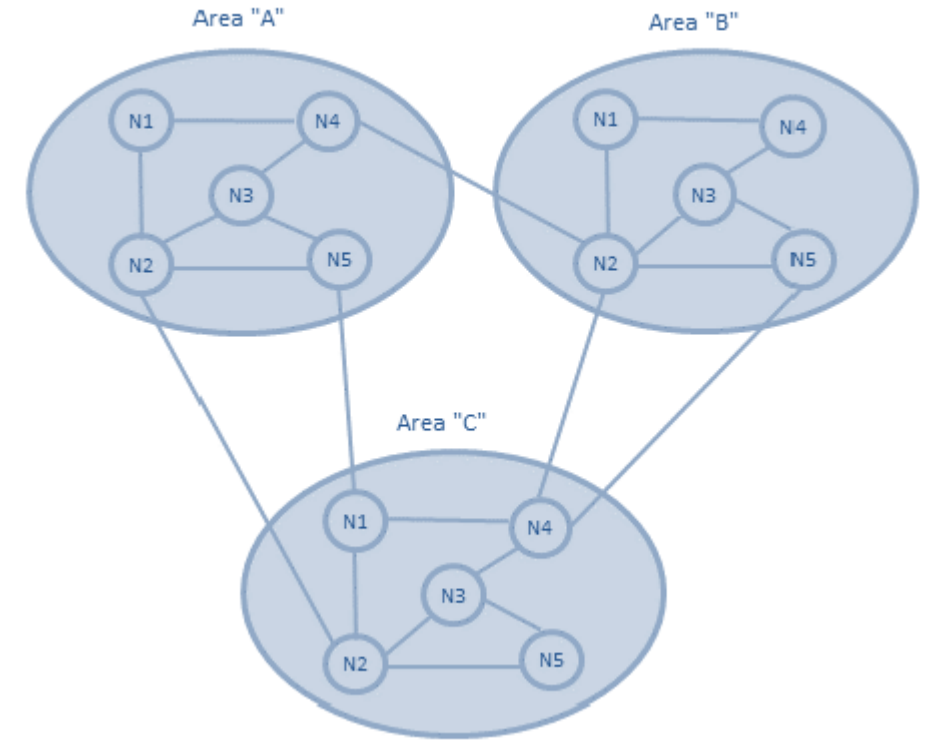
- The Nordic region transitioned from Coordinated NTC (CNTC) to Flow-Based Market Coupling (FBMC)
- FBMC integrates grid physics directly in the market clearing algorithm (Euphemia), maximizing welfare under network constraints
- It uses Power Transfer Distribution Factors (PTDFs) and Generation Shift Keys (GSKs) derived from the Common Grid Model (CGM)
- Market coupling considers the influence of each bidding zone's net position on Critical Network Elements (CNEs)



Source: "Methodology and concepts for the Nordic Flow-Based Market Coupling Approach", Nordic TSOs

From Node-to-CNE to Zone-to-CNE PTDFs

- TSOs compute **nodal** PTDFs from the **linearized DC** power flow equations
- These are aggregated into **zone-to-CNE** PTDFs using **GSKs**, which distribute zonal net position changes among generators and loads
- **Different GSK strategies exist** — based on available capacity, equal participation, or technology mix
- **The choice of the GSK strategy** directly **affects** resulting PTDFs, available capacity, and ultimately **zonal prices**



From Physical Network Representation to Market Coupling

FBMC as a natural extension of SDDP's PTDF-based formulation

- **SDDP** natively represents the power network using **PTDF-based formulations**, so both *nodal and zonal (flow-based)* representations are **naturally supported**.
- **Flow-Based Market Coupling (FBMC)** can therefore be **directly modeled in SDDP** by replacing the nodal PTDF matrix with the zonal one.
- **The bridge** is straightforward

$$PTDF^{AREA} = PTDF^{NODAL} \times GSK$$

Dimensions: (M x A) (M x N) (N x A)

Because SDDP operates in the PTDF domain, representing FBMC or any zonal aggregation is fully consistent and within the model's optimization structure.

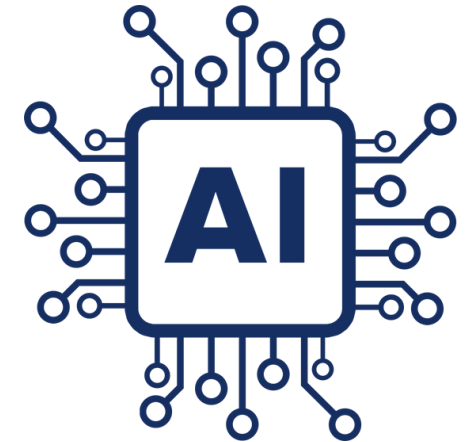
Challenges Beyond the Day-Ahead Horizon

- For mid- and long-term (weeks to months ahead), market players face the main difficulties:
 1. Selecting **representative zonal PTDFs** matching future conditions (depending on the net positions of each time step and scenario)
 2. Estimating **consistent GSKs**, given evolving load, renewable generation, and outages
- The relationship between net positions, grid topology, and binding constraints is highly non-linear and time-dependent
- This makes accurate medium-term price curve modeling particularly complex
- These challenges highlight the need for stochastic, multi-stage models — exactly the domain where SDDP excels

Potential solution: Data-Driven Estimation of PTDFs and GSKs

Using Machine Learning to match PTDFs / GSKs and current condition

- **Train ML models on historical market and system data:**
 1. Inputs: historical net positions, flows, prices, grid states, and zone-to-zone PTDFs
 2. Outputs: best-fitting PTDF and GSK patterns for each system condition
- The model can identify the most probable PTDFs and GSKs for future configurations **based on the past available information**
- This enables as consistent as it can be flow-based price projections even without direct access to future FB parameters



Key takeaways



Conceptual foundation

- **Nodal model:** PTDFs independent of net positions but require full network data and lead to nodal prices.
- **FBMC:** Aggregates nodes into zones using **GSKs**, linking zonal net positions to physical constraints.
- **Trade-off:** Preserves market design consistency but introduces approximation and dependence on GSK definition.



Practical challenges

- **No published PTDFs/GSKs for medium- and long-term horizons** → uncertainty for forward prices, water values, and storage optimization.
- **Extending FBMC beyond day-ahead is cumbersome** — strongly dependent on evolving network conditions.



Possible paths forward

- **Data-driven approaches** can infer PTDF/GSK patterns from historical data, improving forward consistency.
- **Hybrid strategy:**
 - Apply **FBMC constraints** in mid-term stages to capture **market congestion**.
 - **Simplify to zonal or equivalent representation** for **long-term optimization**.

FBMC improves short-term transparency and efficiency compared to CNTC approach, but its long-term application requires hybrid or data-driven approaches to remain tractable and meaningful.



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